

Data Analytics Internship – Task Submission

Customer Demographics Study

A Comprehensive Collection of Charts & Graphs

Submitted By

Anushka Jain

Intern ID: CITS3178

Domain: Data Analytics

Internship Period: 03 June 2026 – 15 July 2026

Duration: 6 Weeks

1. Demographic Segmentation Overview

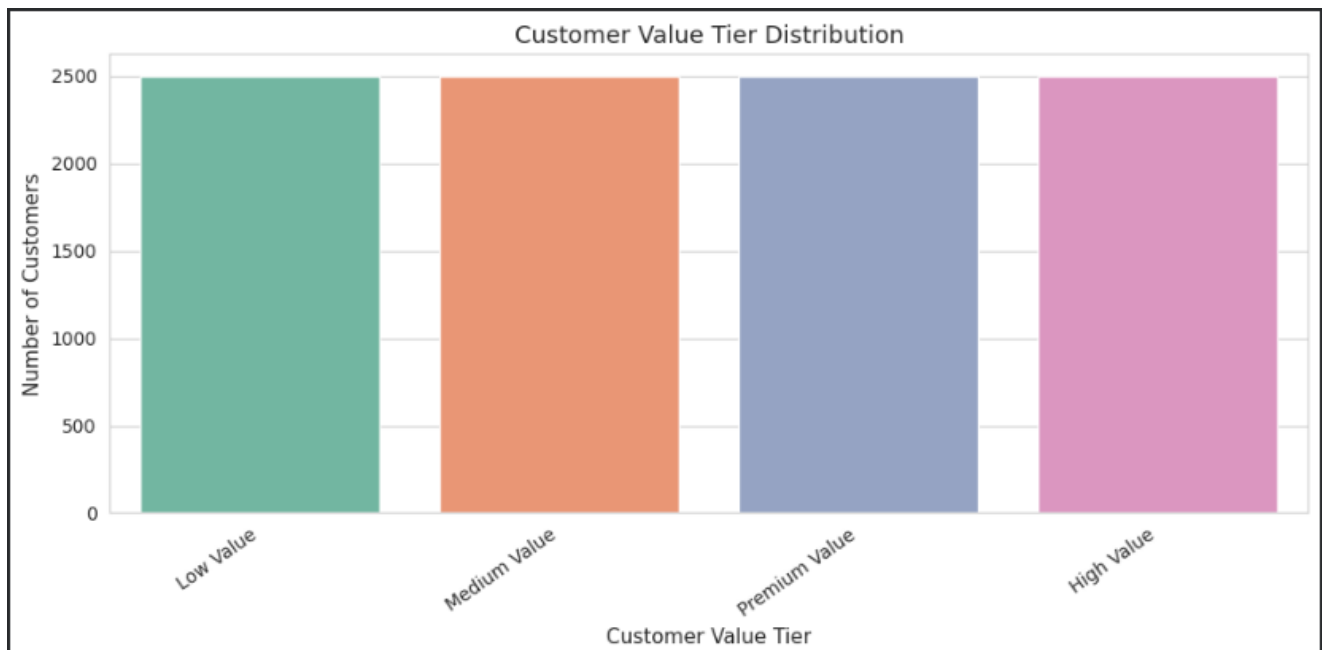


Fig 1.1 – Customer Value Tier Distribution (Bar Chart)

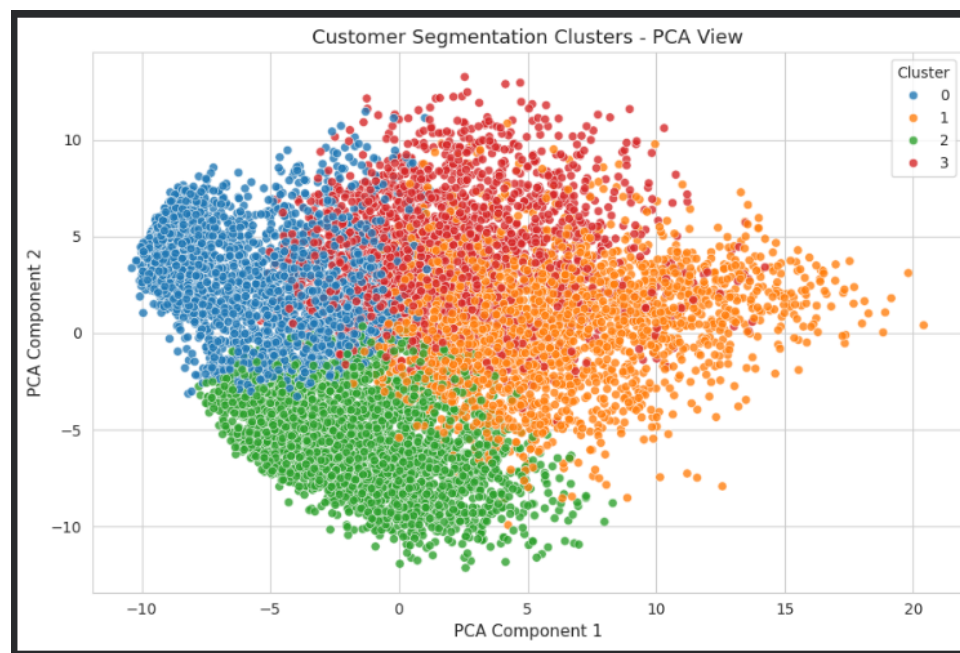


Fig 1.2 – Customer Segmentation Clusters, PCA View (Scatter Plot)

2. Feature Relationships & Key Drivers

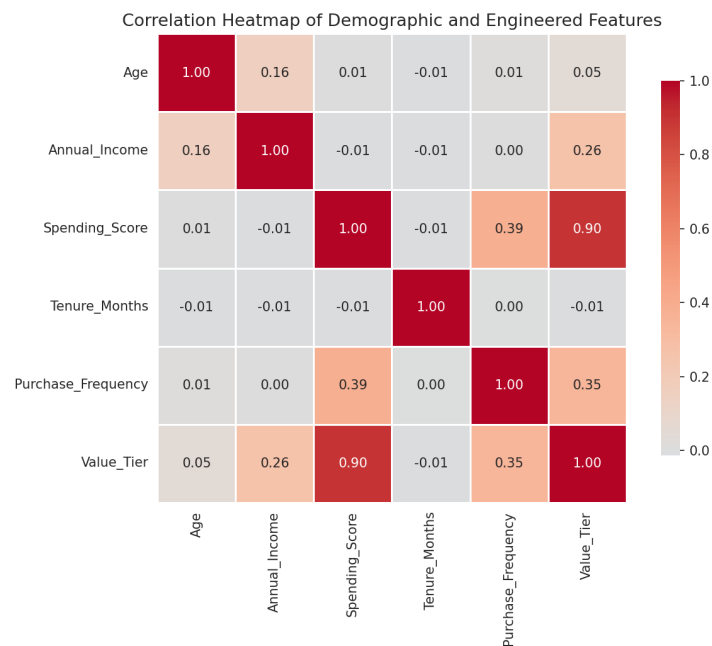


Fig 2.1 – Correlation Heatmap of Demographic and Engineered Features

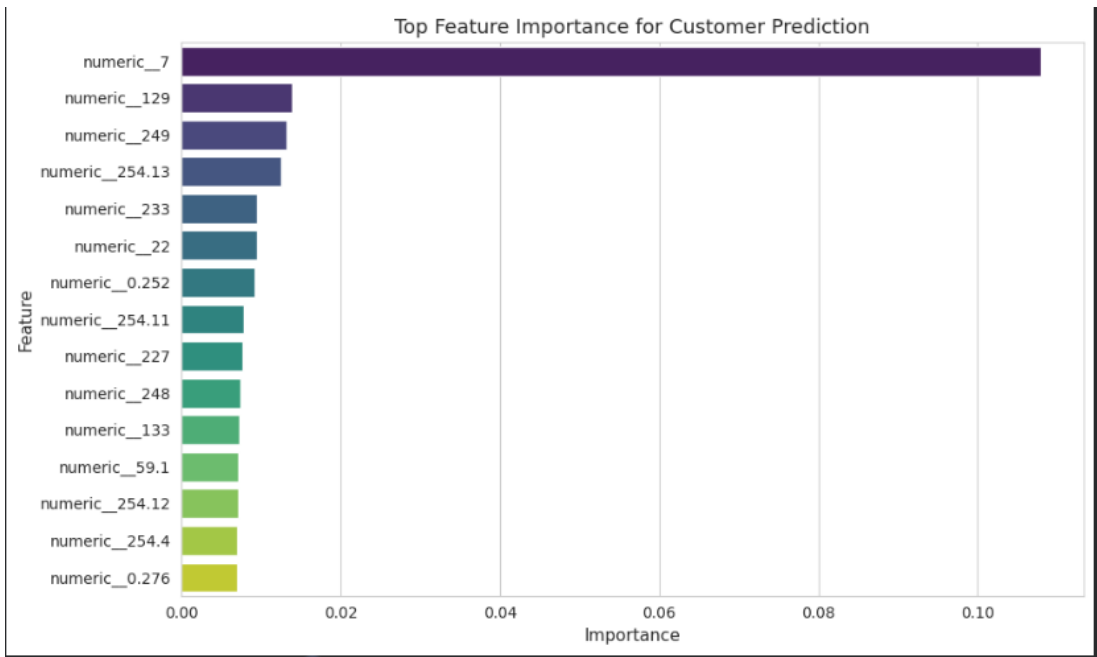


Fig 2.2 – Top Feature Importance for Customer Prediction (Random Forest)

3. Customer Profile Snapshot



Fig 3.1 – Age, Gender, Income, Spending Score, Value Tier & Cluster Distributions (Panel View)

4. Segmentation Diagnostics

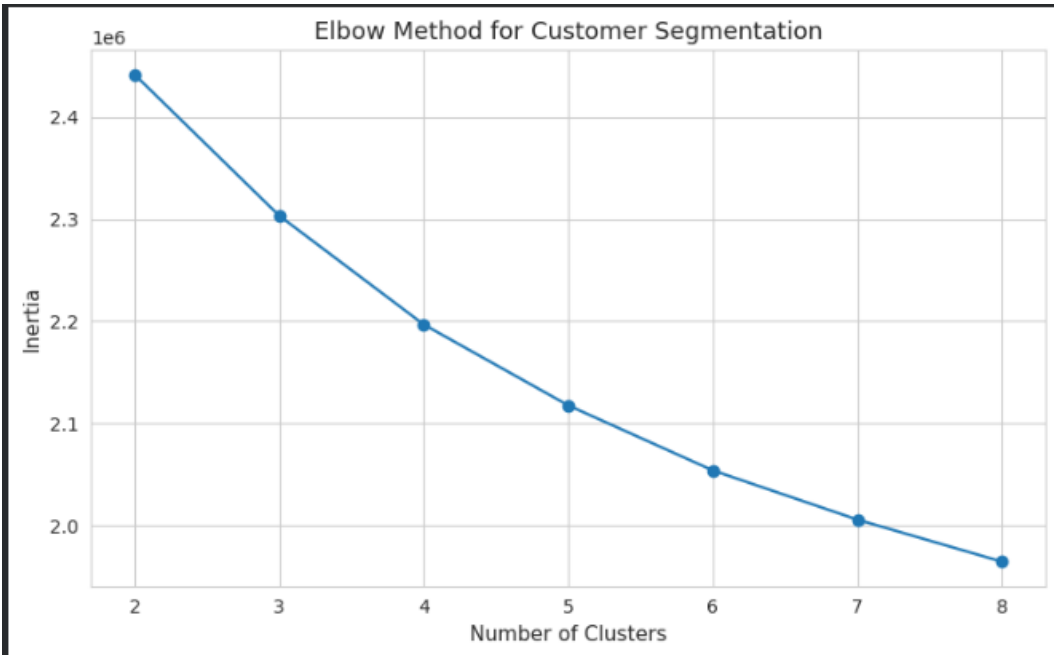


Fig 4.1 – Elbow Method for Customer Segmentation (Inertia vs. Cluster Count)

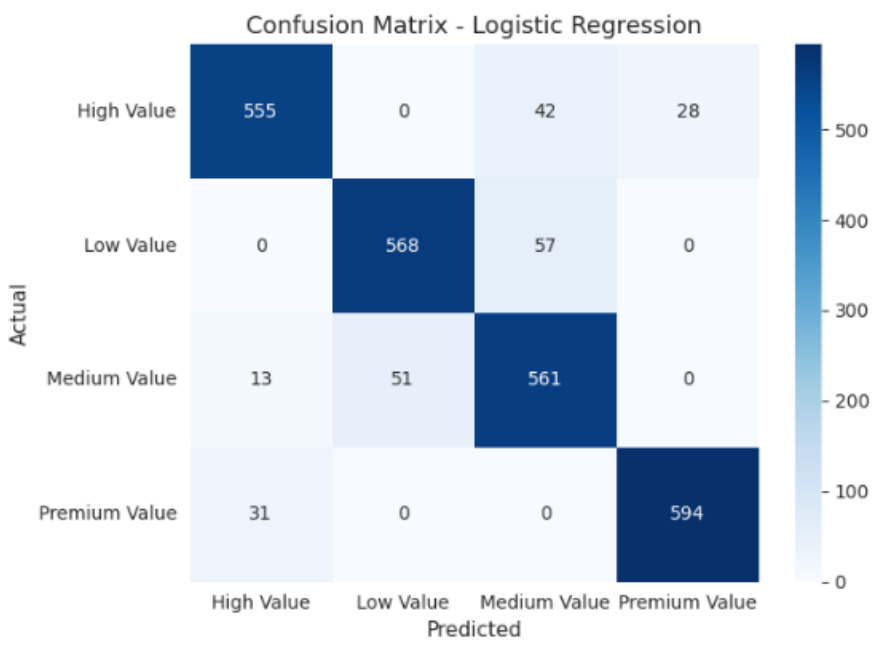


Fig 4.2 – Confusion Matrix – Logistic Regression (Value Tier Classification)

5. Predictive Model Performance

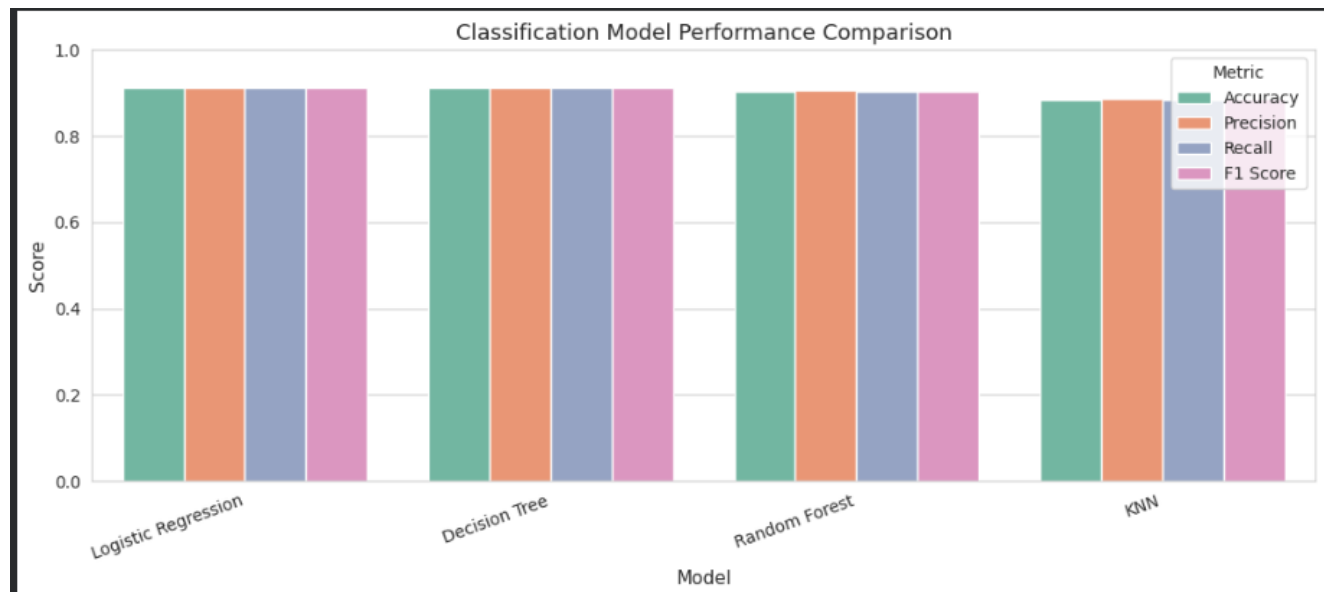


Fig 5.1 – Classification Model Performance Comparison (Accuracy, Precision, Recall, F1 Score)

Summary: The customer base segments cleanly into four roughly even-sized value tiers (Low, Medium, Premium, High), with spending score and purchase frequency emerging as the strongest correlates of value tier. PCA visualization confirms four well-separated behavioral clusters, and the elbow method supports a four-cluster solution as a reasonable balance between model simplicity and explained variance. Logistic Regression and Decision Tree models achieved the highest classification accuracy (~91%) in predicting customer value tier from demographic and behavioral features, with Random Forest close behind and KNN trailing slightly at ~88%.